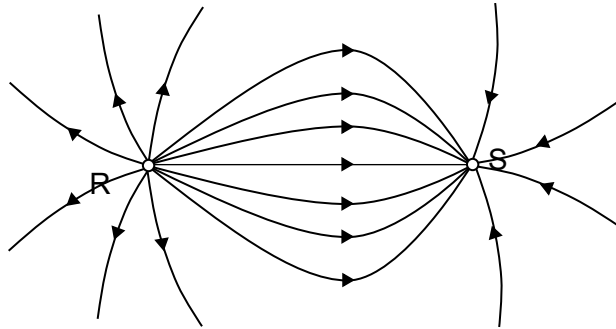


Tutorial 14 Electric Fields Suggested Solutions

D1 (a) (i)



- Field lines at R are more closely spaced.
- Field points from R to S.
- Field lines are not symmetrical about the midpoint of R and S.

(ii) electric field strength due to R,

$$E_R = \frac{Q_R}{4\pi\epsilon_0 r^2} = \frac{7.2 \times 10^{-9}}{4\pi(8.85 \times 10^{-12})(0.060/2)^2} = 7.193 \times 10^4 \text{ N C}^{-1} \text{ towards the right}$$

electric field strength due to S,

$$E_S = \frac{Q_S}{4\pi\epsilon_0 r^2} = \frac{3.6 \times 10^{-9}}{4\pi(8.85 \times 10^{-12})(0.060/2)^2} = 3.597 \times 10^4 \text{ N C}^{-1} \text{ towards the right}$$

resultant electric field strength at mid-point,

$$\begin{aligned} E &= E_R + E_S = 7.193 \times 10^4 + 3.597 \times 10^4 \\ &= 1.079 \times 10^5 = 1.08 \times 10^5 \text{ N C}^{-1} \text{ towards the right} \end{aligned}$$

(iii)

$$F = \frac{Q_R Q_S}{4\pi\epsilon_0 r^2} = \frac{(7.2 \times 10^{-9})(3.6 \times 10^{-9})}{4\pi(8.85 \times 10^{-12})(0.060)^2} = 6.474 \times 10^{-5} = 6.5 \times 10^{-5} \text{ N (shown)}$$

(b) Consider forces acting on S.

horizontal equilibrium:

$$T \sin \theta = F = 6.474 \times 10^{-5} \text{ ----- (1)}$$

vertical equilibrium:

$$T \cos \theta = W = 0.49 \times 10^{-3} \text{ ----- (2)}$$

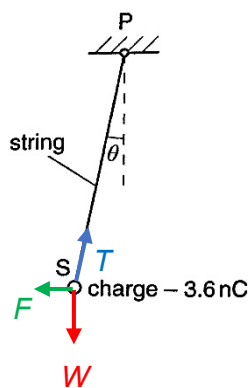
$$\frac{(1)}{(2)} \quad \tan \theta = \frac{6.474 \times 10^{-5}}{0.49 \times 10^{-3}}$$

$$\theta = 7.526 = 7.53^\circ$$

subst. $\theta = 7.526^\circ$ into (2)

$$T \cos 7.526^\circ = 0.49 \times 10^{-3}$$

$$T = 4.943 \times 10^{-4} = 4.94 \times 10^{-4} \text{ N}$$



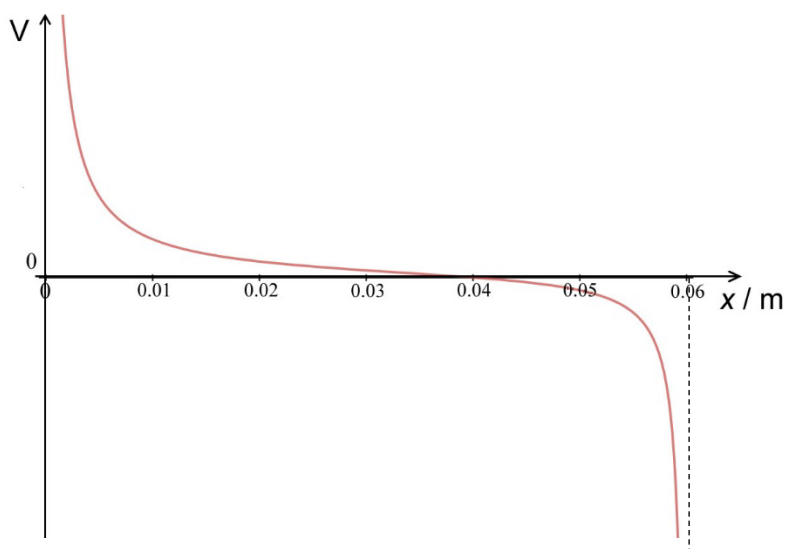
(c)
$$V = V_R + V_S = \frac{Q_R}{4\pi\epsilon_0 x} + \frac{Q_S}{4\pi\epsilon_0 (0.060 - x)}$$

when $V = 0$,

$$-\frac{Q_R}{4\pi\epsilon_0 x} = \frac{Q_S}{4\pi\epsilon_0 (0.060 - x)}$$

$$-\frac{7.2}{x} = \frac{-3.6}{(0.060 - x)}$$

$$x = 0.040 \text{ m}$$



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- D2 (a) (i)** The two regions are within the spheres A and B.

Since A and B are made of conducting material (metal), charges on A and B initially will move under the influence of the electric forces until electrostatic equilibrium is reached. At this point, there is no resultant electric force acting on the charges and the electric fields inside the spheres are zero.

- (ii)** 1. From the diagram, distance within which field strength = 0,
radius of A = 4.00 cm
radius of B = 2.00 cm

2. The spheres have charges of the opposite sign.

According to the x-axis in the graph, direction to the right is positive. Since the resultant electric field strength is positive at all points between the two spheres, it always points from A towards B. This implies that the individual field strengths of A and B are always pointing in the same direction to the right such that their vector sum gives a resultant field strength that always points to the right. Hence, A and B are positively and negatively charged, respectively.

Note: If both A and B are positively charged, then the field strength due to A will point away from A to the right (positive) and the field strength due to B will point away from B to the left (negative). Hence the field strength should be positive when close to A and negative when close to B. Furthermore, somewhere between A and B there would be a point at which the resultant field strength is zero since the individual field strengths point in opposite directions at all points between A and B. (Similar explanation applies if both A and B are negatively charged.)

- (iii)** 1. energy gained by nucleus
= work done by electric force
= electric force $F \times$ distance moved d
= charge $q \times$ average $E \times d$
= $(3 \times 1.60 \times 10^{-19})(1.2 \times 10^5)(0.050)$
= 2.88×10^{-15} J

2. By Newton's second law, resultant electric force

$$F = ma$$

$$a = \frac{F}{m} = \frac{qE}{m} = \frac{(3 \times 1.60 \times 10^{-19})(3.0 \times 10^5)}{(7 \times 1.66 \times 10^{-27})} = 1.239 \times 10^{13} = 1.24 \times 10^{13} \text{ m s}^{-2}$$

3. From Newton's second law, the acceleration a of the nucleus is given by $a = \frac{F}{m} = \frac{qE}{m}$. Thus, a is directly proportional to the resultant electric field strength E .

Its acceleration thus decreases in a similar manner as E from a high value at $x = 4.0$ cm to a low but non-zero value at around $x = 18.5$ cm before increasing again to the initial value at $x = 28.0$ cm.

- D3 (a) (i)** The electric field strength at a point is the negative gradient of the potential-distance function at that point. Mathematically, $E = -\frac{dV}{dx}$ where x is the distance from the source charge.

- (ii) 1.** At $x = 25.0$ cm, the potential gradient is negative. Thus the electric field strength at that point is in the positive x -direction, pointing from A to B.

OR

The electric field strength at $x = 25.0$ cm points in the positive x -direction, pointing from A to B since electric field strength points in the direction of decreasing potential.

- 2.** At $x = 35.0$ cm, the potential gradient is zero. Hence, $E = -\frac{dV}{dx} = 0$.

Force on electron placed at that point, $F = eE = 0$ N.

- 3.** Since metal is a conducting material, charges are able to move freely. Any presence of an electric field, due to either external charges or charges within the metal sphere itself, will cause the excess charges to redistribute themselves onto the surface of the sphere until the electric field within the sphere is zero, at which point the collective movement of the charges will cease. Since $E = 0 = -\frac{dV}{dx}$, this implies that the potential gradient is zero within the sphere, and the potential graph is a horizontal line.

- (c) (i)** At $x = 15.0$ cm, $V = 515$ V $\Rightarrow Vx = 7725$

At $x = 20.0$ cm, $V = 405$ V $\Rightarrow Vx = 8100$

Since the Vx values at $x = 15.0$ cm and $x = 20.0$ cm differ significantly, the student's statement is incorrect.

- (ii) The potential graph in Fig. 4.2 shows the total potential, which is a summation of the individual potentials due to A and B:

$$V = \frac{Q_A}{4\pi\epsilon_0 x} + \frac{Q_B}{4\pi\epsilon_0 (0.500 - x)}.$$

The product Vx is not constant, and is a function of x .

Note: For a single charge, electric potential $V = \frac{Q}{4\pi\epsilon_0 x}$, where x is the distance from the charge, $Vx = \frac{Q}{4\pi\epsilon_0}$ is indeed a constant.

- (d) (i) By the conservation of energy,
 increase in K.E. of electron = decrease in E.P.E. of electron

$$\frac{1}{2}mv^2 - 0 = 0 - (-eV)$$

minimum speed v occurs when potential V is minimum (at $x = 35$ cm)

$$\frac{1}{2}m_e v_{\min}^2 = eV_{\min}$$

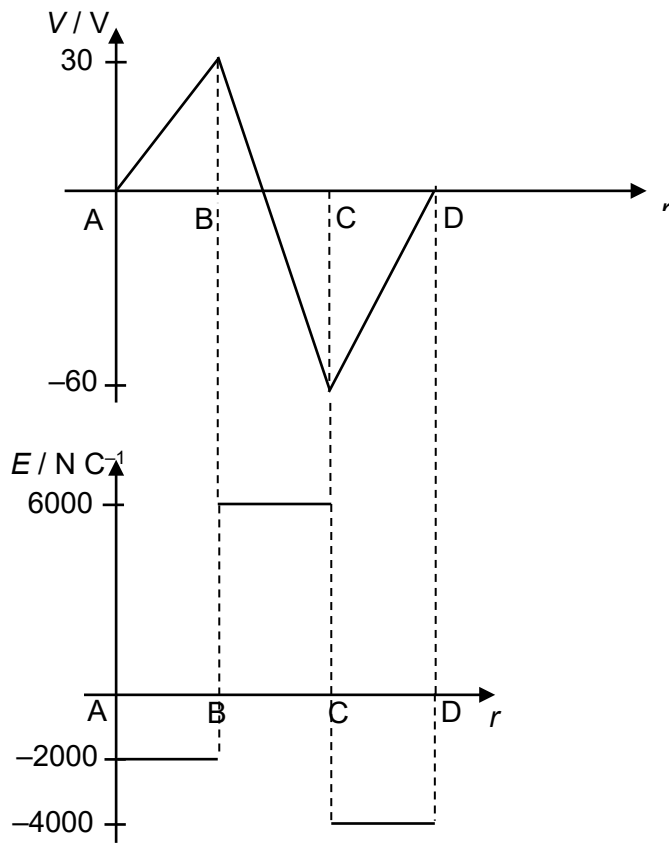
$$v_{\min} = \sqrt{\frac{2eV_{\min}}{m_e}} = \sqrt{\frac{2(1.60 \times 10^{-19})(295)}{9.11 \times 10^{-31}}} = 1.018 \times 10^7 = 1.02 \times 10^7 \text{ m s}^{-1}$$

- (ii) The path of such an electron will be a curve passing through this point where $x = 35$ cm.

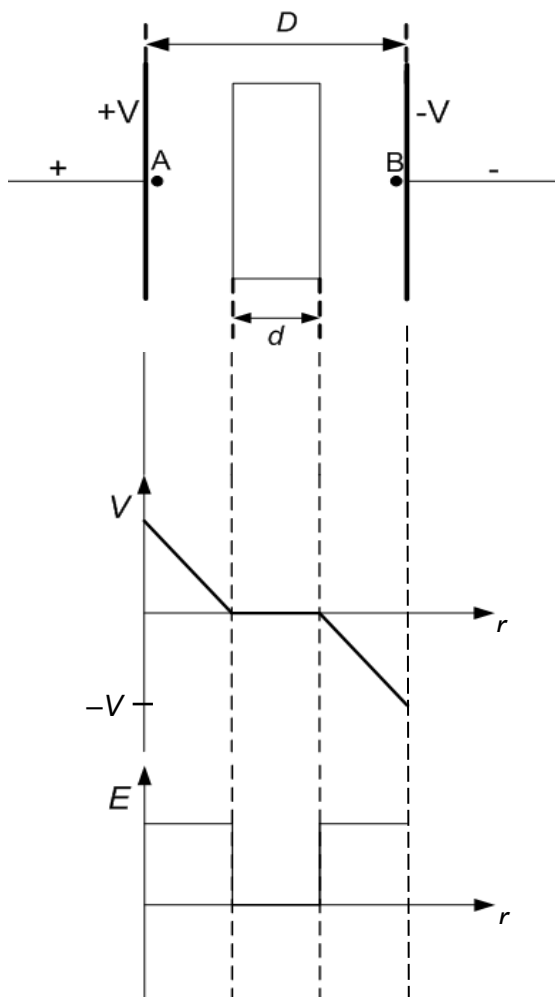
Note:

- As the electron approaches this point ($x = 35$ cm) between the two spheres, its acceleration decreases in magnitude, becomes momentarily zero at that point (resultant field strength = 0 at turning point of potential graph), and increases as it moves away. Hence, if the electron approaches the point in a curved path, the path should be “flatter” around $x = 35$ cm.
- The possible paths of the electron can be traced out in the reverse direction. It can be projected from $x = 35$ cm with the speed as calculated in the previous part. Such an electron, when reaching infinity, will have zero speed as required. The direction of projection, on the other hand, can be arbitrarily chosen. If the electron is projected along the line joining the two spheres, the path will be a straight line; otherwise, the path will be a curve.

D4



D5



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D6 (a) (i) From the diagram, the plates have a separation of 2.5 cm.

$$|E| = \left| -\frac{dV}{dx} \right| = \frac{(+600) - (-600)}{2.5 \times 10^{-2}} = 48000 \text{ V m}^{-1}$$

(ii) $W_{\text{ext}} = \Delta U = q\Delta V = (8.0 \times 10^{-19})[(500) - (-400)] = 7.2 \times 10^{-16} \text{ J}$

Note: Since a positive charge is being brought from a lower (A) to a higher (B) potential, positive work is done by the external force to increase its electric potential energy, which is consistent with the numerical answer.

- (b)**
- Since the electric field is uniform between the plates, the potential gradient between the plates is constant.
 - The potential difference between the centre line and either plate will be the same.
 - As the potentials of both plates have the same magnitude but opposite sign, the centre line will be at zero potential.

Note: For points on the centre line but outside the two plates, by symmetry (of the pattern of the field lines), the potential difference between the centre line the either plate is still the same. Hence, the same argument applies.

D7 (a) $F_E = qE = (1.60 \times 10^{-19})(2.7 \times 10^5) = 4.32 \times 10^{-14} \text{ N}$

(b) (i) $W = Fd = (4.32 \times 10^{-14})(0.078) = 3.37 \times 10^{-15} \text{ J}.$

Note: The electric force does positive work (the proton accelerates from A to B) as the constant electric force (weight is neglected) on the proton is acting vertically downwards, and displacement is also downwards.

- (ii)** By the principle of conservation of energy,
work done by the electric field (decrease in E.P.E.) = increase in K.E. of the proton.

Therefore, the gain in K.E. = $3.37 \times 10^{-15} \text{ J}$

(iii) WD by field = decrease in E.P.E. = $q(V_A - V_B)$

$$3.37 \times 10^{-15} = q\Delta V$$

$$\Delta V = V_A - V_B = \frac{3.37 \times 10^{-15}}{1.60 \times 10^{-19}} = 21.1 \text{ kV}$$

Since the field lines points downwards and fields lines always point in the direction of decreasing potential, point A is at a higher potential.

(Note also that since $V_A - V_B > 0$, $V_A > V_B$)

D8 (a) (i)

$$F_E = ma$$

$$q \frac{\Delta V}{d} = m_e a$$

$$\left(1.60 \times 10^{-19}\right) \frac{45 - 0}{6.0 \times 10^{-3}} = \left(9.11 \times 10^{-31}\right) a$$

$$a = 1.32 \times 10^{15} \text{ m s}^{-2}$$

The direction of the acceleration is vertically downwards.

Since the electric field is uniform and pointing vertically upwards between the parallel plates, the electric force, which is the only force acting on the electron, is constantly acting vertically downwards. Hence acceleration of the electron due to the electric force is also vertically downwards.

Note: Conclusion cannot be drawn based on the observation of the trajectory, since the actual form of trajectory (parabolic) was not explicitly stated in the question (although it can be deduced from the information provided). One thus has to reason based on the uniformity and direction of the electric field.

(ii) Draw the field lines vertically upwards and equally spaced between the plates (at least 5-6 lines should be drawn).

(iii) Since the electron only experiences a constant vertical downward electric force, its horizontal velocity remains uniform. Thus, the path it takes is parabolic (just like projectile motion in a uniform gravitational field).

(b) Take displacement upwards and to the right as positive.

Consider the motion of the electron from A to B:

Vertically

$$a_y = -1.32 \times 10^{15} \text{ m s}^{-2}$$

$$u_y = u \sin 15^\circ$$

$$s_y = 0$$

Horizontally

$$a_x = 0$$

$$u_x = u \cos 15^\circ$$

$$s_x = 25 \times 10^{-3} \text{ m}$$

$$t = \frac{s_x}{u_x} = \frac{25 \times 10^{-3}}{u \cos 15^\circ} \Rightarrow \text{sub into } s_y = u_y t + \frac{1}{2} a_y t^2$$

$$0 = u \sin 15^\circ \left(\frac{25 \times 10^{-3}}{u \cos 15^\circ} \right) + \frac{1}{2} (-1.32 \times 10^{15}) \left(\frac{25 \times 10^{-3}}{u \cos 15^\circ} \right)^2$$

$$u = 8.12 \times 10^6 \text{ m s}^{-1}$$

Since K.E. is converted to E.P.E. and then back to K.E., assuming no energy losses, the speed at which the electron leaves slit B is the same as that when it enters slit A.

D9 (a) (i) $Q = CV = 200 \times 10^{-6} \times 30 = 6.0 \times 10^{-3} \text{ C}$

(ii) $U = \frac{1}{2}CV^2 = \frac{1}{2} \times 200 \times 10^{-6} \times 30^2 = 0.090 \text{ J}$

(b) (i) The total charge

(ii) The potential difference across each capacitor.

(c) Since charge is conserved

$$Q_i = Q_f = Q_{200,f} + Q_{100,f} = 6.0 \times 10^{-3} \text{ C}$$

Potential difference across each capacitor is the same

$$V_{100} = V_{200}$$

$$\frac{Q_{100,f}}{C_{100}} = \frac{Q_{200,f}}{C_{200}}$$

$$Q_{100,f} = \frac{100 \times 10^{-6}}{200 \times 10^{-6}} (6.0 \times 10^{-3} - Q_{100,f})$$

$$Q_{100,f} = 2.0 \times 10^{-3} \text{ C}$$

$$V_{100} = \frac{Q_{100,f}}{C_{100}} = \frac{2.0 \times 10^{-3}}{100 \times 10^{-6}} = 20 \text{ V}$$

Total energy stored by the two capacitors

$$U_{total,f} = \frac{1}{2}V^2(C_{100} + C_{200}) = \frac{1}{2} \times 20^2 \times (100 + 200) \times 10^{-6} = 0.060 \text{ J}$$

Note: Total energy stored by both capacitors is less than the energy stored initially by the 200 μF capacitor.

D10 (a) (i) $Q = CV = 500 \times 10^{-12} \times 200 = 1.0 \times 10^{-7} \text{ C}$

(ii) $U = \frac{1}{2}CV^2 = \frac{1}{2} \times 500 \times 10^{-12} \times 200^2 = 1.0 \times 10^{-5} \text{ J}$

(b) (i) Since charge is conserved

$$Q_f = Q_i$$

$$V_f = \frac{Q_i}{C_f} = \frac{1.0 \times 10^{-7}}{100 \times 10^{-12}} = 1.0 \times 10^3 \text{ V}$$

(ii) By the conservation of energy,

$$\begin{aligned}
 E_{\text{initial}} &= E_{\text{final}} \\
 U_{\text{initial}} + WD &= U_{\text{final}} \\
 WD &= \frac{1}{2}Q(V_f - V_i) \\
 &= \frac{1}{2} \times 1.0 \times 10^{-7} (1000 - 200) \\
 &= 4.0 \times 10^5 \text{ J}
 \end{aligned}$$

C1 (a) The first term is the electric potential energy due to the Coulomb attraction of the two ions while the second term arises from the repulsion of the orbital electrons surrounding the nuclei of the two ions. The exponent of r in the first term is -1 because the force of attraction between the two ions varies as $1/r^2$.

(b)
$$F = -\frac{dU}{dr} = \frac{-e^2}{4\pi\epsilon_0} \left(\frac{r^{-2}}{-1} \right) + (-9C)(r^{-10})$$

At equilibrium, $F = 0 \quad \therefore \frac{e^2}{4\pi\epsilon_0 r_0^2} = \frac{9C}{r_0^{10}} \Rightarrow C = \frac{e^2 r_0^8}{36\pi\epsilon_0}$

(c) Substituting the expression for C into the equation U ,

$$U_{\text{min}} = \frac{-e^2}{4\pi\epsilon_0 r_0} + \frac{e^2 r_0^8}{36\pi\epsilon_0 r_0^9} = \frac{e^2}{\pi\epsilon_0 r_0} \left(\frac{1}{36} - \frac{1}{4} \right) = -\frac{2}{9} \frac{e^2}{\pi\epsilon_0 r_0}$$

(d) The energy required to break the bond between the ions is

$$|U| = \frac{2(1.6 \times 10^{-19})^2}{9\pi(8.85 \times 10^{-12})(0.3 \times 10^{-9})} = 6.8 \times 10^{-19} \text{ J}$$